Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech | *Subject: Control Systems* | *Time: 03 Hours*
Code: 100502 | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) A control system with feedback generally has
(i) Lower stability (ii) Reduced steady-state error
(iii) Increased nonlinearity (iv) Higher sensitivity to parameter variations

(b) In signal flow graph analysis, a loop in contact with another loop is called
(i) Forward path (ii) Self-loop
(iii) Non-touching loop (iv) Touching loop

(c) For a second-order underdamped system, increasing damping ratio decreases
(i) Rise time only (ii) Peak overshoot
(iii) Settling time only (iv) Natural frequency

(d) The number of branches in a root locus equals the number of
(i) Zeros (ii) Closed-loop poles
(iii) Open-loop poles (iv) Gain constants

(e) Breakaway points in root locus occur where
(i) Gain becomes zero (ii) Multiple roots exist
(iii) Poles cross imaginary axis (iv) Characteristic equation changes sign

(f) A Type-1 system has zero steady-state error for
(i) Step input (ii) Ramp input
(iii) Parabolic input (iv) Impulse input

(g) The slope of magnitude plot for a single pole in Bode plot is
(i) +20 dB/decade (ii) –20 dB/decade
(iii) –40 dB/decade (iv) +40 dB/decade

(h) Phase margin is the additional phase lag required to make the system
(i) Critically damped (ii) Oscillatory
(iii) Unstable (iv) Faster

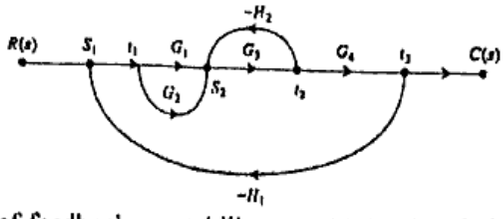
(i) Root-locus based controller design mainly involves adjustment of
(i) Pole-zero locations (ii) Sampling frequency
(iii) Input amplitude (iv) State variables

(j) State-space analysis is particularly useful for
(i) Nonlinear algebraic equations (ii) Single-input single-output systems only
(iii) Multi-variable systems (iv) Frequency response only

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Q.2 (a) Explain signal flow graphs and derive Mason's Gain Formula. Determine the transfer function $(\frac{C(s)}{R(s)})$ for given signal flow graph. [7]



(b) Discuss the effects of feedback on stability, sensitivity, bandwidth, and steady-state error of a control system. [7]

Q.3 (a) A unity feedback system has: [7]

$$G)S) = \frac{25}{S(S+5)}$$

Determine: (i) Damping ratio (ii) Natural frequency (iii) Peak overshoot (iv) Settling time.

(b) Derive expressions for rise time and maximum overshoot for a standard second-order system. [7]

Q.4 (a) Using the Routh–Hurwitz criterion, determine the stability of the system whose characteristic equation is: $S^4 + 2S^3 + S^2 + 2S + 1 = 0$. [7]

(b) Sketch the root locus for: [7]

$$G(S)H(S) = \frac{K}{S(S+2)(S+4)}$$

Determine the breakaway points and imaginary-axis crossings.

Q.5 (a) Explain the effect of adding poles and zeros on the transient response and stability of a control system. [7]

(b) State and explain the Nyquist stability criterion. Also draw the Nyquist Plots for: [7]

$$G(S)H(S) = \frac{3}{(S+2)(S-1)}$$

Q.6 (a) Discuss gain margin and phase margin and explain their significance in relative stability analysis. [7]

(b) Explain lag compensation and discuss its effect on steady-state accuracy and bandwidth. [7]

Q.7 (a) Discuss the concept of robustness in control systems and explain how controller design improves robustness. [7]

(b) Explain the concepts of state variables, state equations, and state-space representation with suitable example. [7]

Q.8 (a) Discuss the stability of discrete-time systems and explain the role of eigenvalues in determining system stability. [7]

(b) Test the controllability and observability of the system: [7]

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} C = [1\ 0]$$

Q.9 Write short notes on ***any two*** of the following: [7 x 2 = 14]

(a) Routh–Hurwitz stability criterion and its significance
(b) Gain margin and phase margin in frequency response analysis
(c) Initial value theorem and final value theorem
(d) closed-loop control systems

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